



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
Faculty of Engineering, Built Environment and
Information Technology

EBB 320

CONTROL SYSTEMS

PRACTICAL 1: HARDWARE IMPLEMENTATION AND SYSTEM IDENTIFICATION

Name and Surname	Student Number	Signature	% Contribution
Hanré Coetzer	21507334		33
Cornelis Claasen	24704611		33
B.J. Mncube	23700816		33

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1 Theoretical Background

The accuracy of the estimated transfer function models was evaluated using the Normalized Root Mean Square Error (NRMSE) fit percentage. This metric quantifies how well the simulated model response matches the measured physical data, and is defined as:

$$\text{Fit (\%)} = 100 \times \left(1 - \frac{\|y - \hat{y}\|}{\|y - \bar{y}\|} \right) \quad (1)$$

where y is the measured output data, $\hat{y}$ is the simulated model output, $\bar{y}$ is the mean of the measured output data, and $\|\cdot\|$ denotes the Euclidean norm.

2 Design and Simulation

The previous First Order Plus Dead Time (FOPDT) models made using Matlab's `tfest` function were used to construct the simulation models used in this practical.

$$G_{11} = \frac{0.1175}{s + 0.007592} \quad (2)$$

$$G_{12} = \frac{0.05949}{s + 0.0060685} \quad (3)$$

Matlab's `compare` function was then used to confirm G_{11} 's accuracy. It was found to be $\approx 94\%$ accurate.

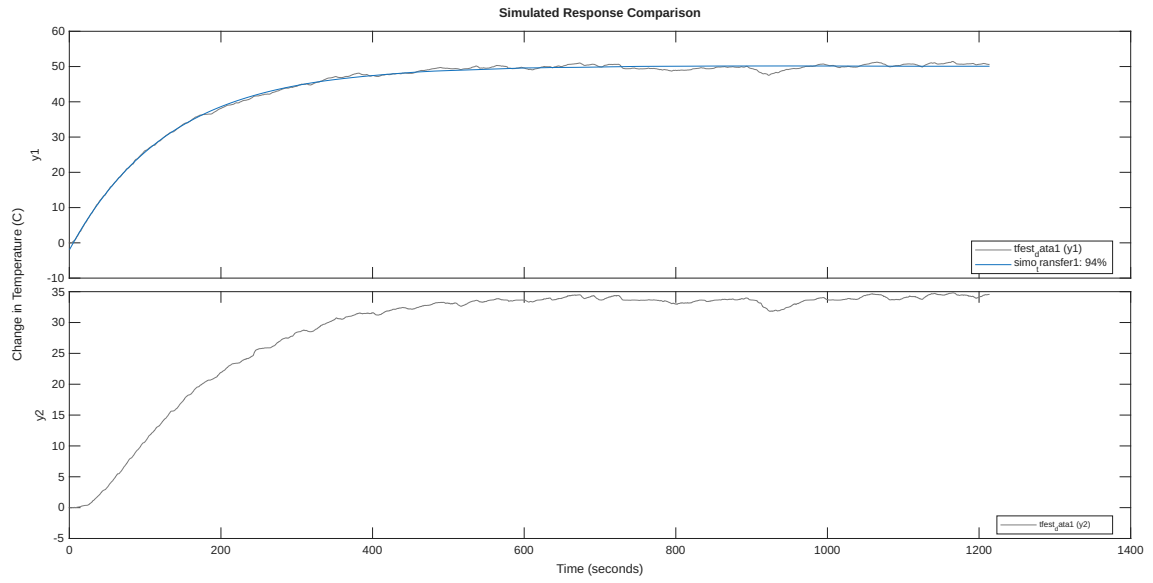


Figure 1: Accuracy comparison of G_{11} 's transfer function

2.1 P Controller

The Proportional (P) controller was designed using MATLAB's `pidtune` function applied to the transfer function G_{11} identified in Practical 1. The resulting controller has the form $C(s) = K_p$, where the proportional gain was tuned to achieve a phase margin of 60° . The P controller produces a fast initial response but introduces a steady-state error because, without an integral term, the controller output falls to zero as the error approaches zero.

Listing 1: P Controller Design (MATLAB)

```
[C1_p, info] = pidtune(G11, 'P');
```

The tuned gain and step-response metrics for the closed-loop system $T(s) = \frac{C_P G_{11}}{1 + C_P G_{11}}$ are summarised in tables 1 and 2.

Table 1: P Controller Tuned Parameters

Parameter	Value
K_p	0.847505
K_i	—
K_d	—
Achieved Phase Margin	60.0°

Table 2: P Controller Step Response Metrics (2% Settling Criterion)

Step Response Parameter	Closed Loop	Open Loop
Overshoot (%)	7.42	0.0
Rise Time (s)	7.9	291.2
Peak Time (s)	21.2	970.5
Settling Time (s)	30.3	524.4
Steady-State Gain	0.9296	1.0

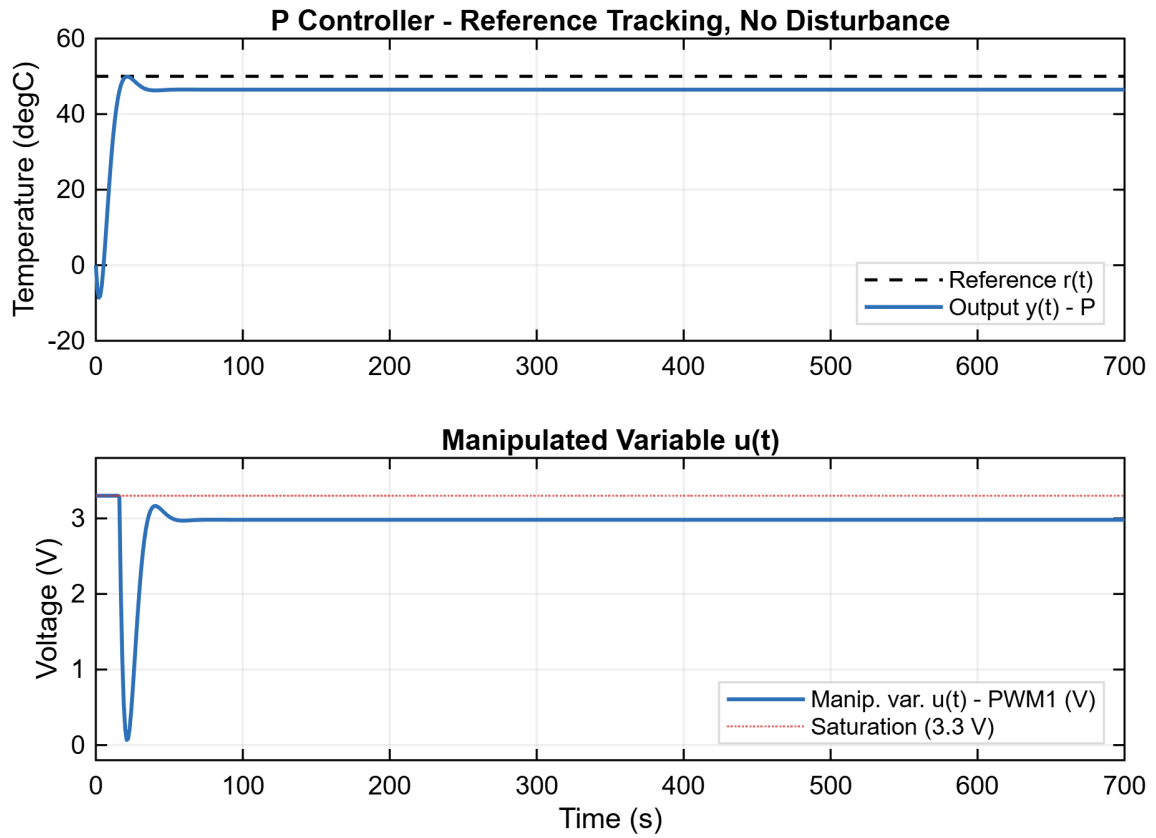


Figure 2: Closed-loop response of the P controller with no disturbance. Top: reference trajectory and controlled variable (temperature). Bottom: manipulated variable (PWM1 voltage).

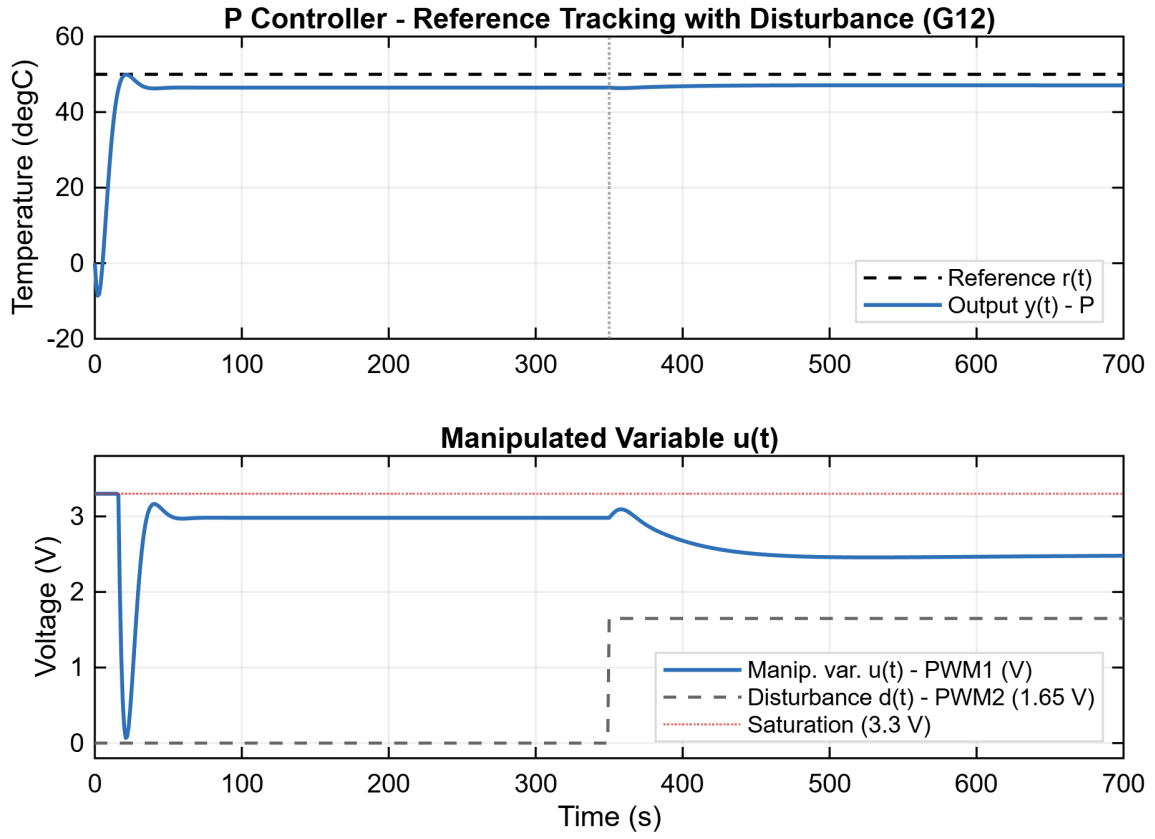


Figure 3: Closed-loop response of the P controller with disturbance via G_{12} . Top: reference and controlled variable. Bottom: manipulated variable (PWM1) and disturbance input (PWM2).

2.2 PI Controller

The Proportional-Integral (PI) controller was designed by adding an integral term to eliminate the steady-state error observed in the P controller. The controller has the form $C(s) = K_p + \frac{K_i}{s}$, which ensures that the integral term accumulates the error over time and drives the steady-state error to zero. The trade-off is a slower response and increased overshoot compared to the P controller.

Listing 2: PI Controller Design (MATLAB)

```
[C1_pi, info] = pidtune(G11, 'PI');
```

The tuned gains and step-response metrics for the closed-loop system are summarised in tables 3 and 4.

Table 3: PI Controller Tuned Parameters

Parameter	Value
K_p	0.075342
K_i	0.001432
K_d	—
Achieved Phase Margin	60.0°

Table 4: PI Controller Step Response Metrics (2% Settling Criterion)

Step Response Parameter	Closed Loop	Open Loop
Overshoot (%)	12.95	0.0
Rise Time (s)	103.1	291.2
Peak Time (s)	232.6	970.5
Settling Time (s)	394.8	524.4
Steady-State Gain	1.0000	1.0

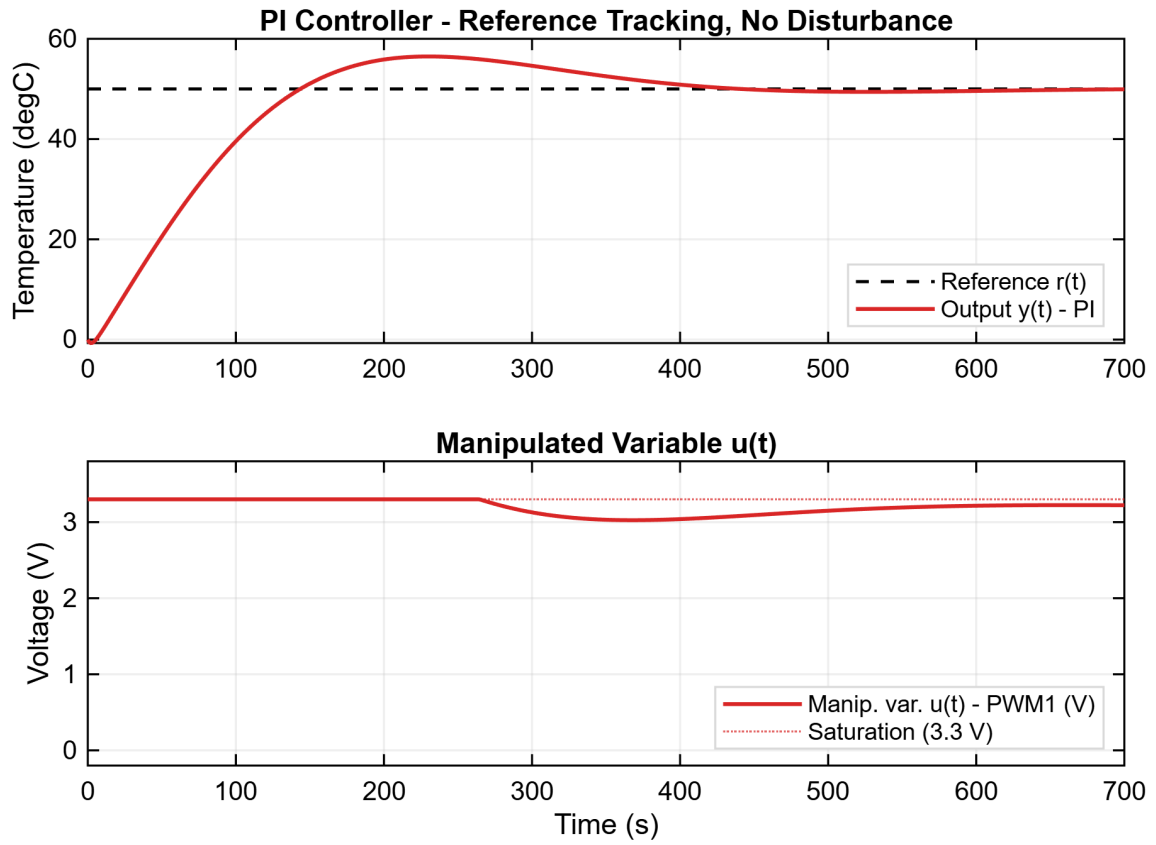


Figure 4: Closed-loop response of the PI controller with no disturbance. Top: reference trajectory and controlled variable (temperature). Bottom: manipulated variable (PWM1 voltage).

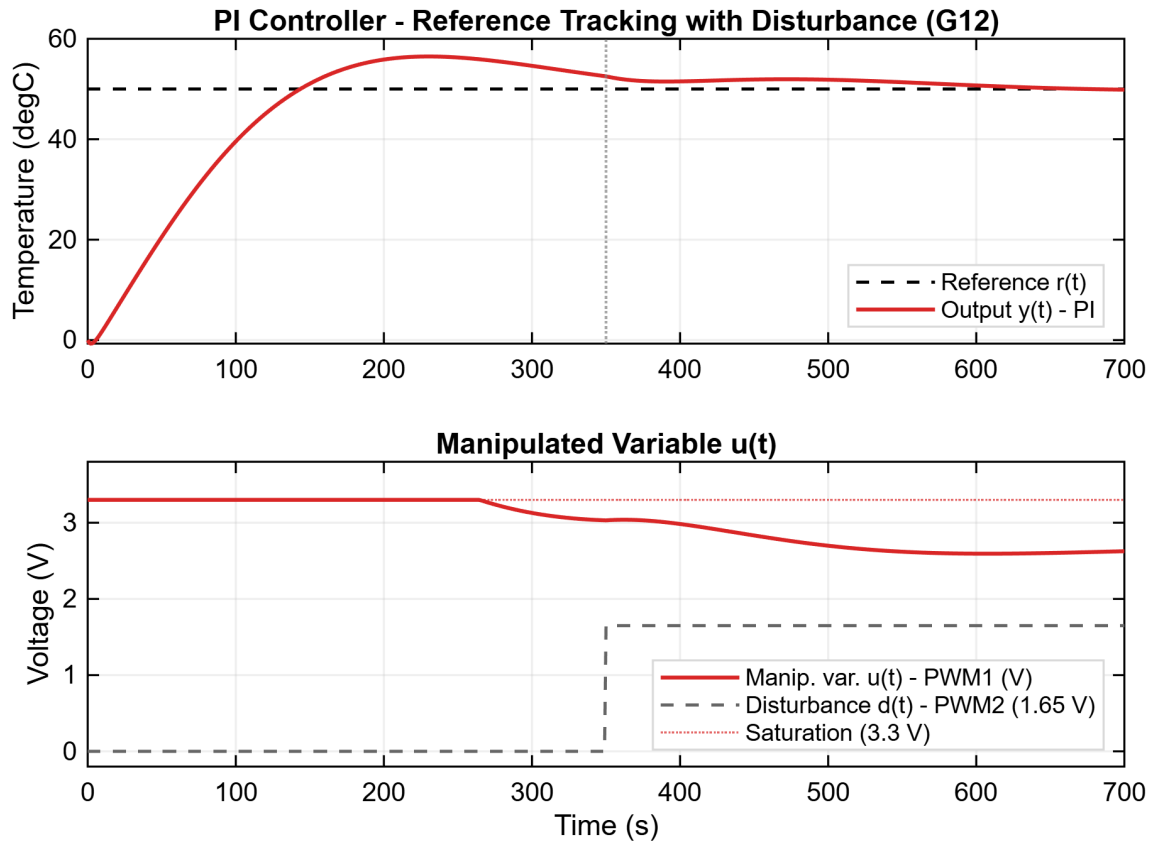


Figure 5: Closed-loop response of the PI controller with disturbance via G_{12} . Top: reference and controlled variable. Bottom: manipulated variable (PWM1) and disturbance input (PWM2).

2.3 PD Controller

The PID parameters were found using Matlab's `piddtune` function with G_{11} as the input. Those parameters were: $K_p = 4.58$, $K_i = 0.0$, and $K_d = 0.0$. The controller was then simulated using the Simulink model below.

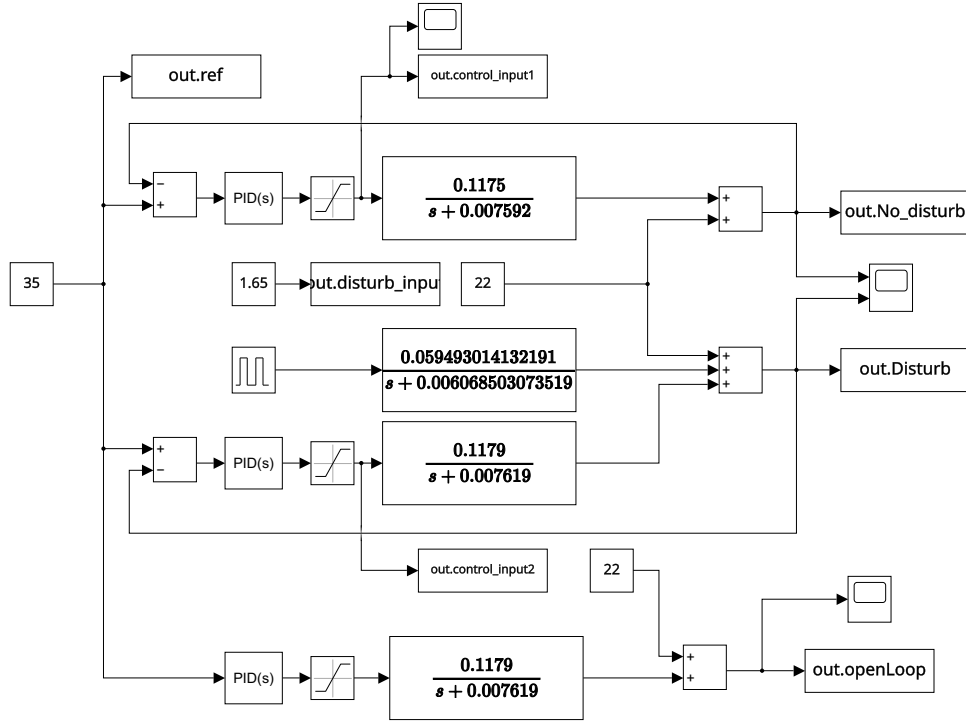


Figure 6: Simulink Model used for PD Controller

The resulting model was then used to find the following step response parameters using Matlab's `stepinfo` function.

Table 5: Table comparing closed loop PD step response parameters to the open loop controller

Step Response Parameter	Closed Loop	Open Loop
Overshoot (%)	8.8047×10^{-5}	0.0
Rise Time (s)	77.3698	251.4783
Peak Time (s)	119.1159	1500
Settling Time (s)	97.0706	446.5085

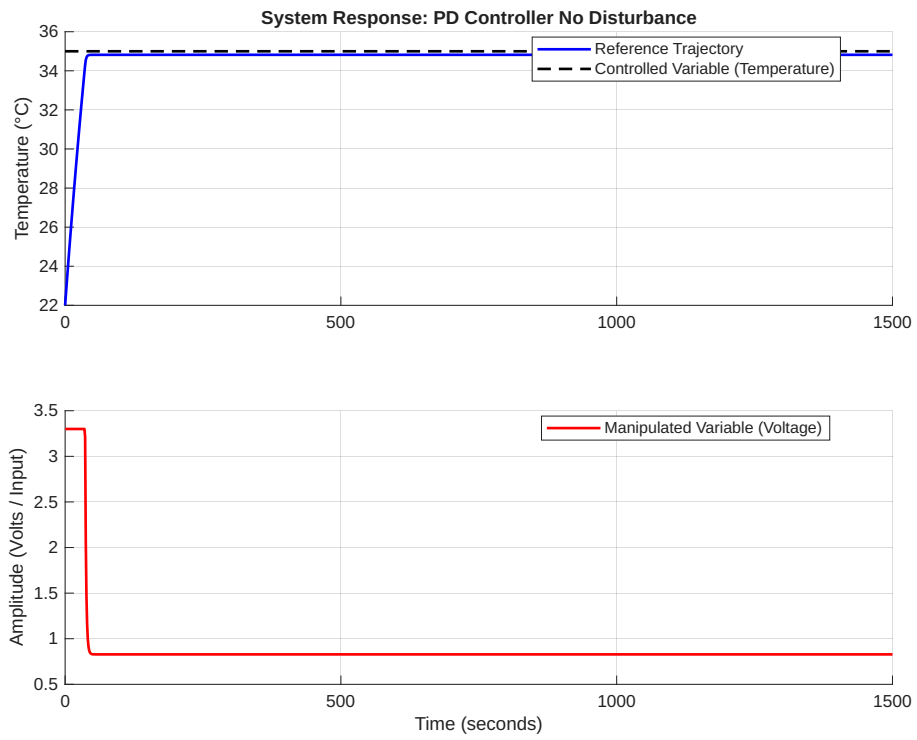


Figure 7: PD Control with no disturbance

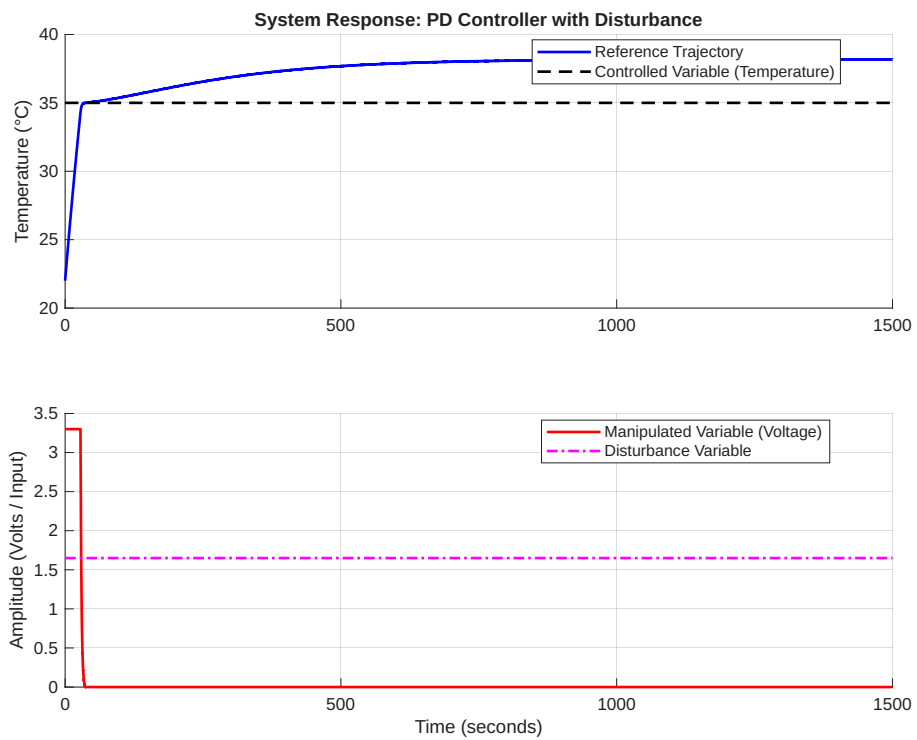


Figure 8: PD Control with disturbance

2.4 PID - Reference Tracking Controller

The parameters found using Matlab's `pidtune` function were: $K_p = 0.112$, $K_i = 0.000935$, and $K_d = 0.0$.

The same Simulink model used for the PD controller[6] was used for the PID reference tracking model.

The following step response parameters where then found using the model:

Table 6: Table comparing closed loop PID controller tuned for reference tracking against the open loop controller

Step Response Parameter	Closed Loop	Open Loop
Overshoot (%)	0.1592	0.0
Rise Time (s)	123.6591	251.4783
Peak Time (s)	432.7289	1500
Settling Time (s)	220.6166	446.5085

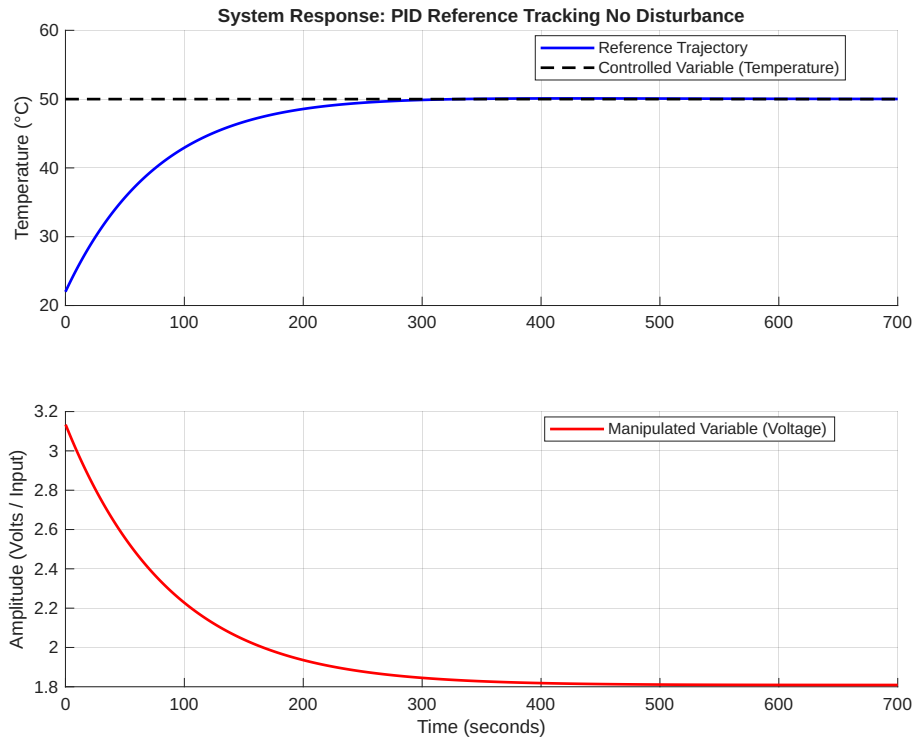


Figure 9: PID reference tracking control with no disturbance

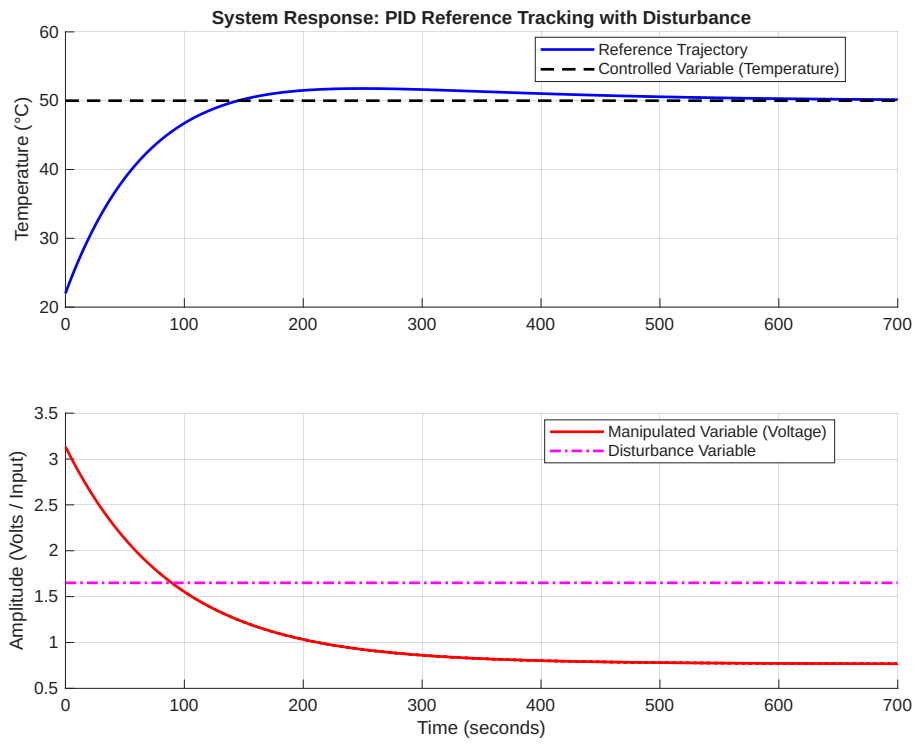


Figure 10: PID reference tracking control with disturbance

2.5 PID - Disturbance Rejection Controller

The parameters used for the disturbance rejection tuning are $K_p = 0.067424$, $K_i = 0.0016631$, and $K_d = 0.68337$.

The controller was then simulated using the Simulink Model below.

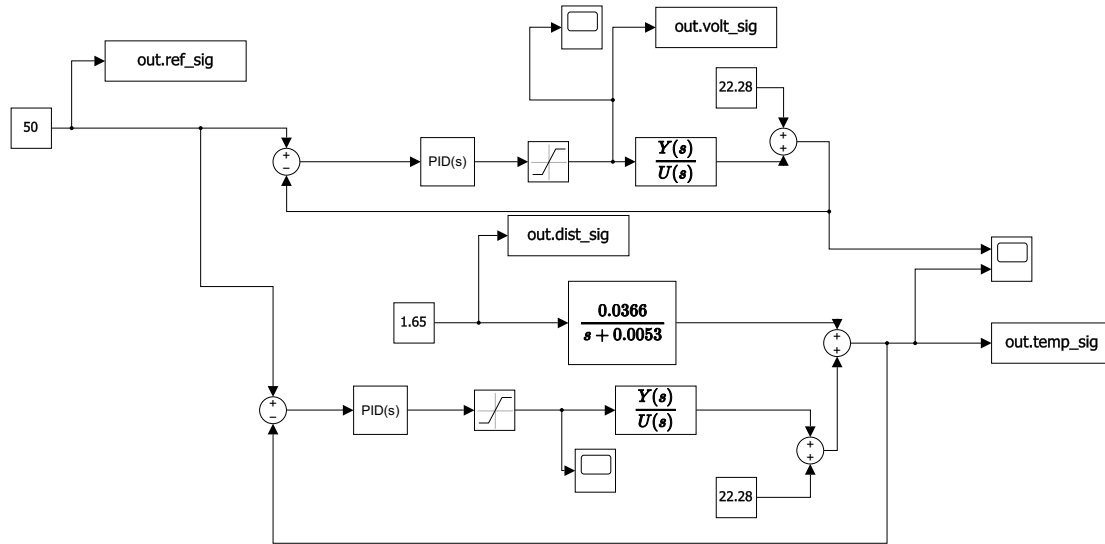


Figure 11: Simulink Model used for PID Disturbance Rejection Controller

Again Matlab's `stepinfo` function was then used to find the following step response parameters.

Step Response Parameter	Closed Loop	Open Loop
Overshoot (%)	9.26	0.0
Rise Time (s)	102.934	251.4783
Peak Time (s)	225.625	1500
Settling Time (s)	366.913	446.5085

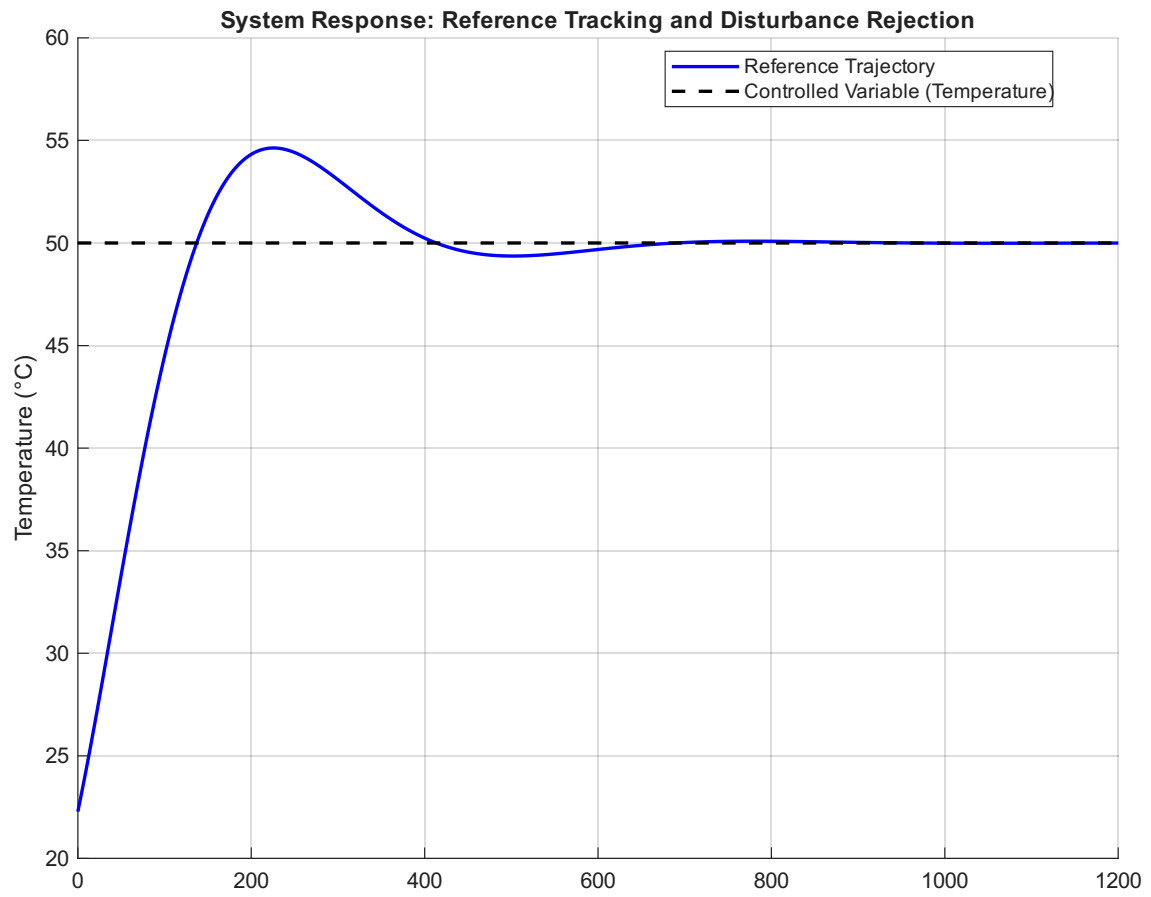
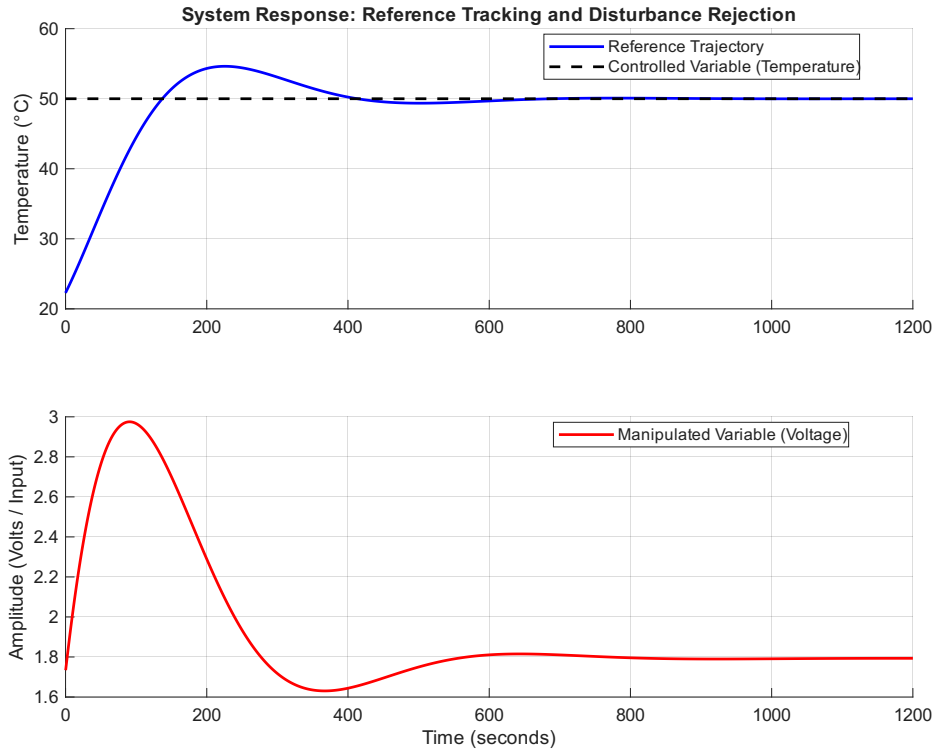
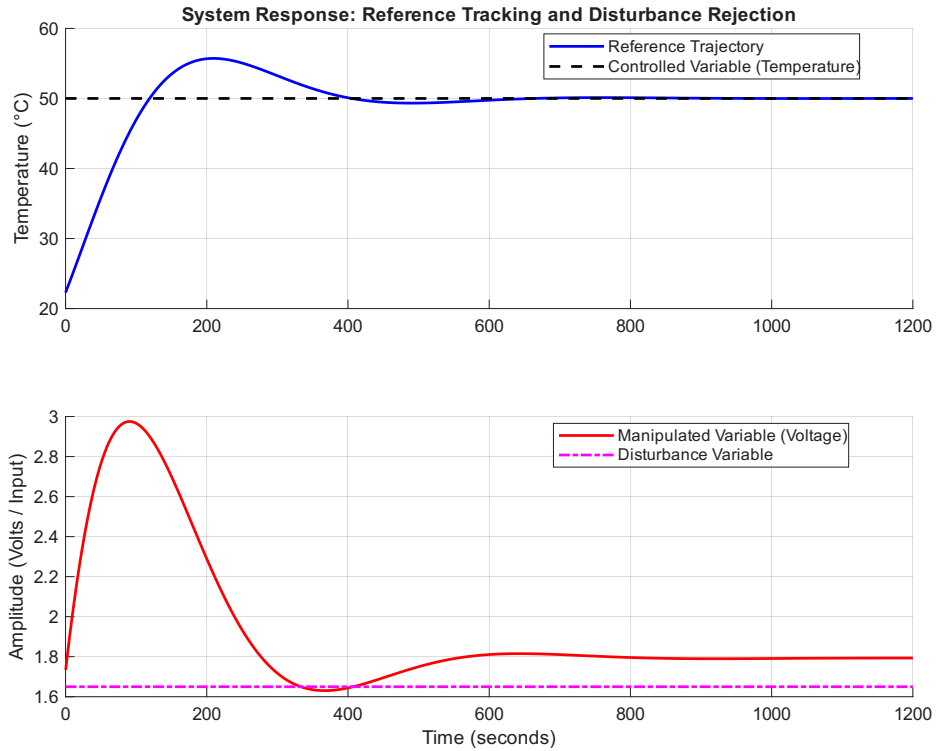


Figure 12: Closed Loop response of PID Disturbance Rejection Controller



(a) Closed-loop response (No Disturbance).



(b) Closed-loop response (50% PWM Disturbance).

Figure 13: Simulink closed-loop step response for the disturbance-rejection PID controller. The bottom figure demonstrates the controller compensating for the external heat disturbance caused by transistor 2.

3 Implementation

3.1 PID Control on MCU

The continuous-time PID controller designed in Simulink was discretized for implementation on the ESP32 microcontroller. The software was developed in C++ and relies on a fixed sample time (Δt) to ensure the integral and derivative calculations remain mathematically accurate.

3.1.1 Discrete PID Algorithm and Hardware Protection

Because the physical plant is constrained by a 0–3.3V operating range mapped to a 0–100% PWM duty cycle, several engineering protections were implemented in the firmware:

- **PID output limiting:** The `pid_out` variable is constrained to the physical limit of 3.3V.
- **Anti-derivative kick:** The first derivative calculation will spike to infinity causing unpredictable behavior, this is handled by implementing a `first_run` flag variable that sets the previous error equal to the current error when the derivative is calculated for the first time.
- **Anti-integral windup:** The integral term does not get updated if the system demands an output higher than the ESP’s limit and the error is not negative. The inverse case is handled in the same way.

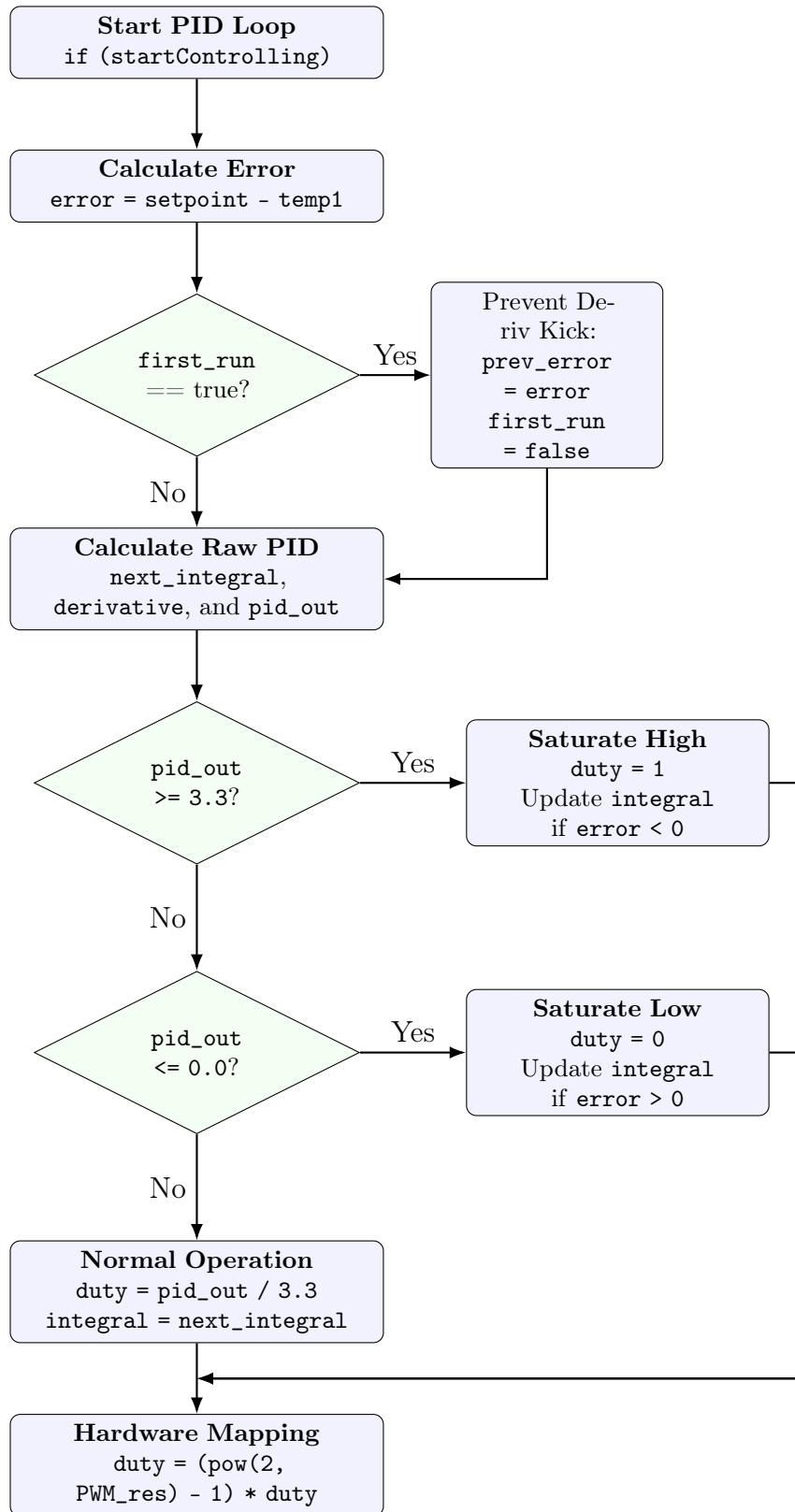


Figure 14: Software execution flow of the discrete PID algorithm, highlighting first-run derivative protection and integral anti-windup logic.

4 Results

4.1 PID Reference Tracking

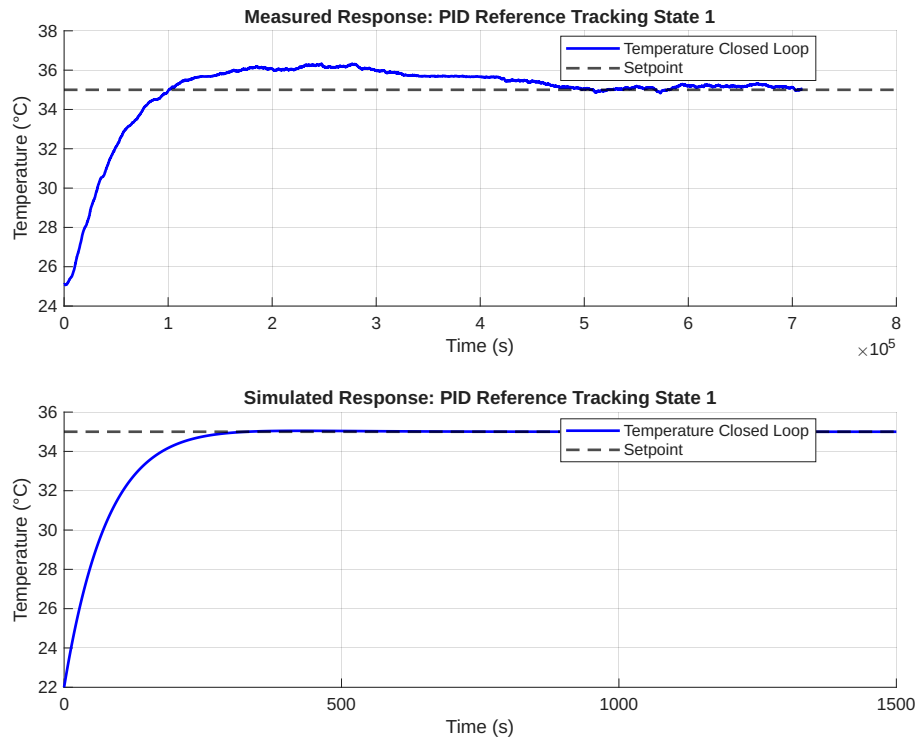


Figure 15: Measured Results for PID reference tracking for state 1

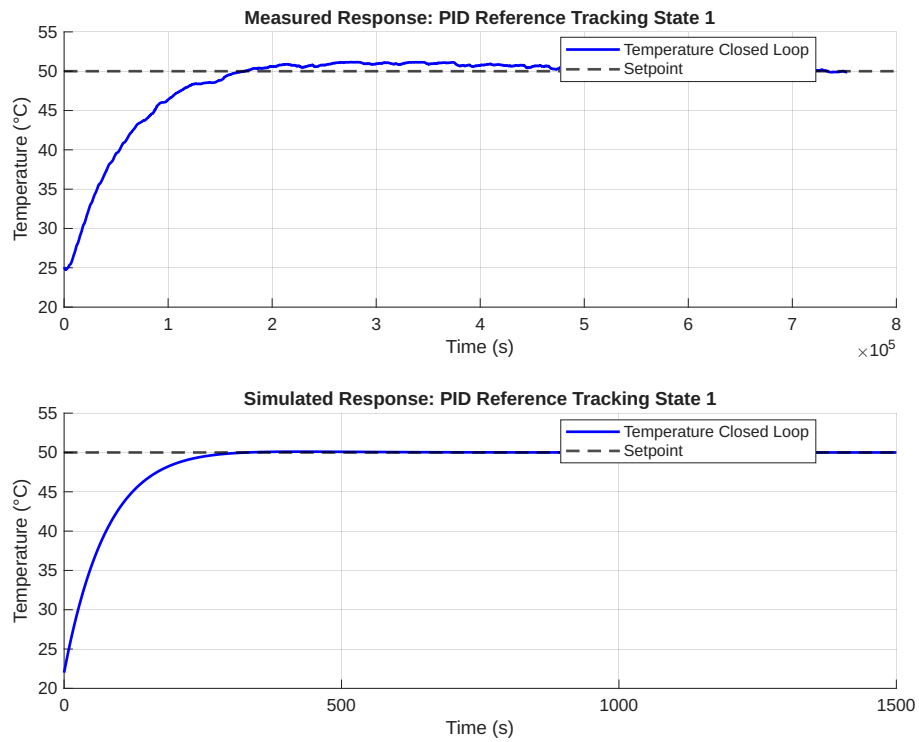


Figure 16: Measured Results for PID reference tracking for state 2

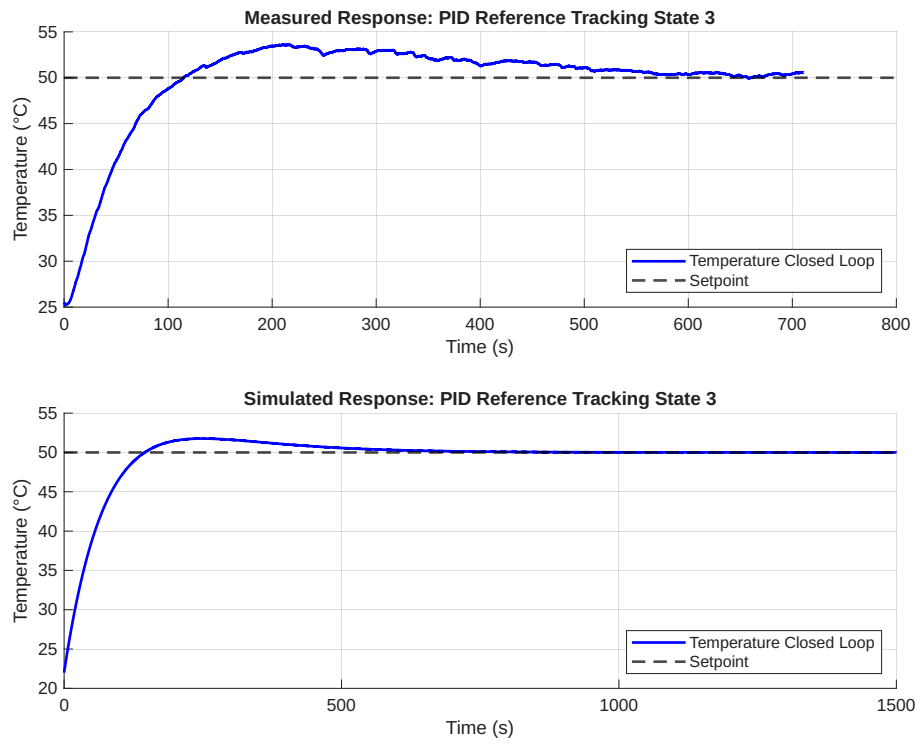


Figure 17: Measured Results for PID reference tracking for state 3

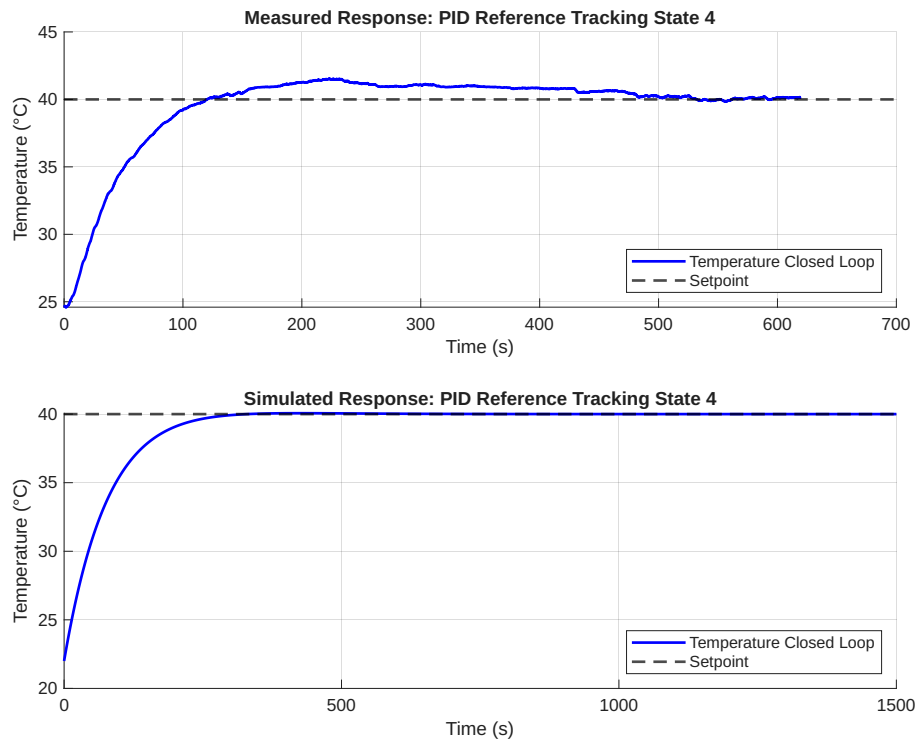


Figure 18: Measured Results for PID reference tracking for state 4

4.2 PID Disturbance Rejection

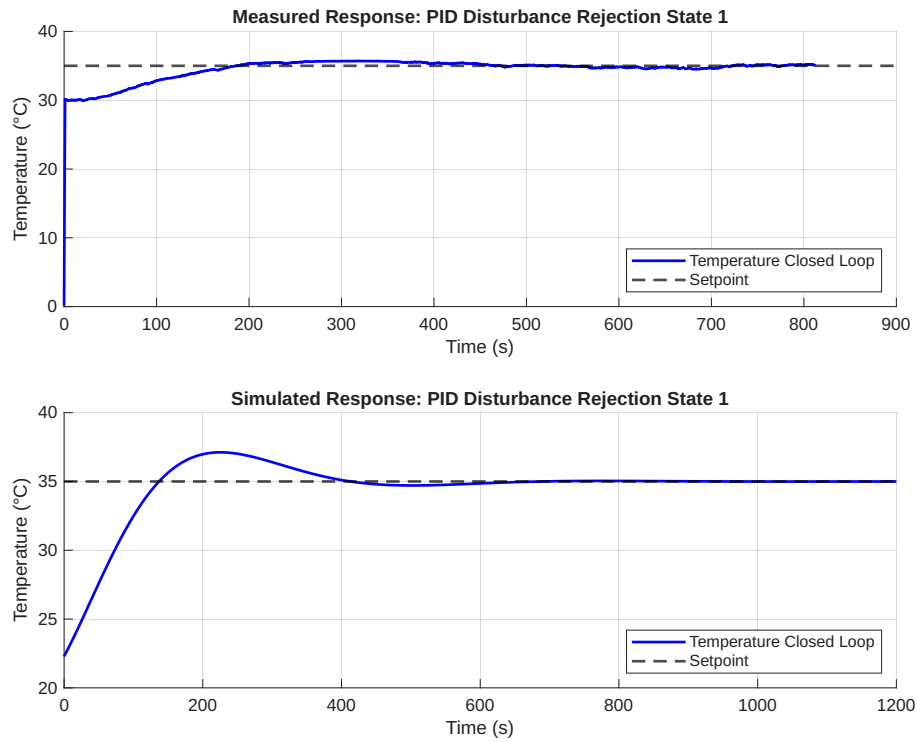


Figure 19: Measured Results for PID disturbance rejection tracking for state 1

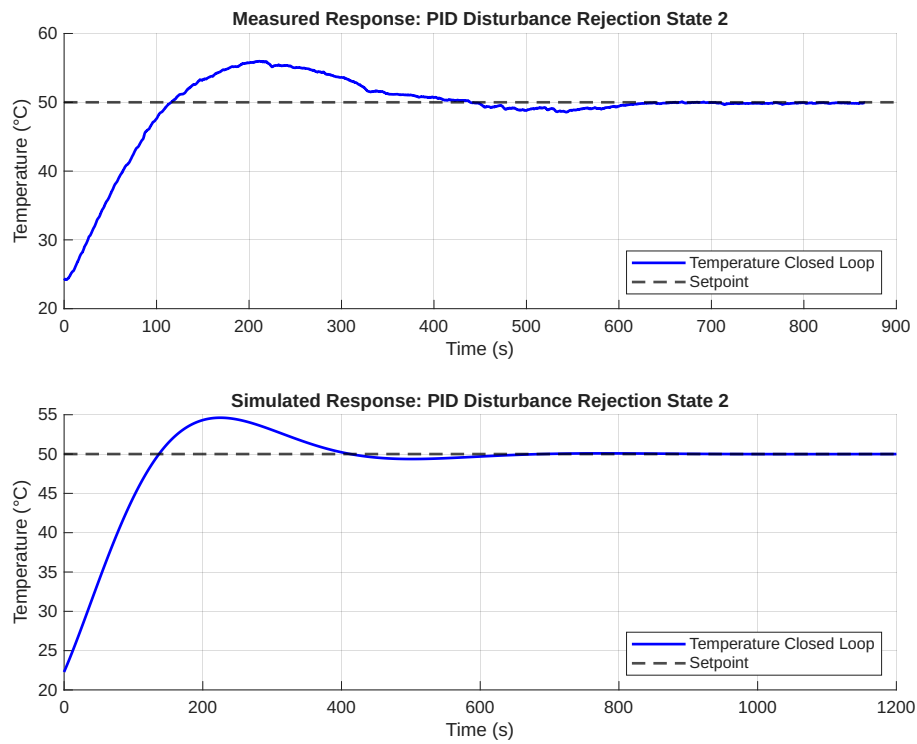


Figure 20: Measured Results for PID disturbance rejection tracking for state 2

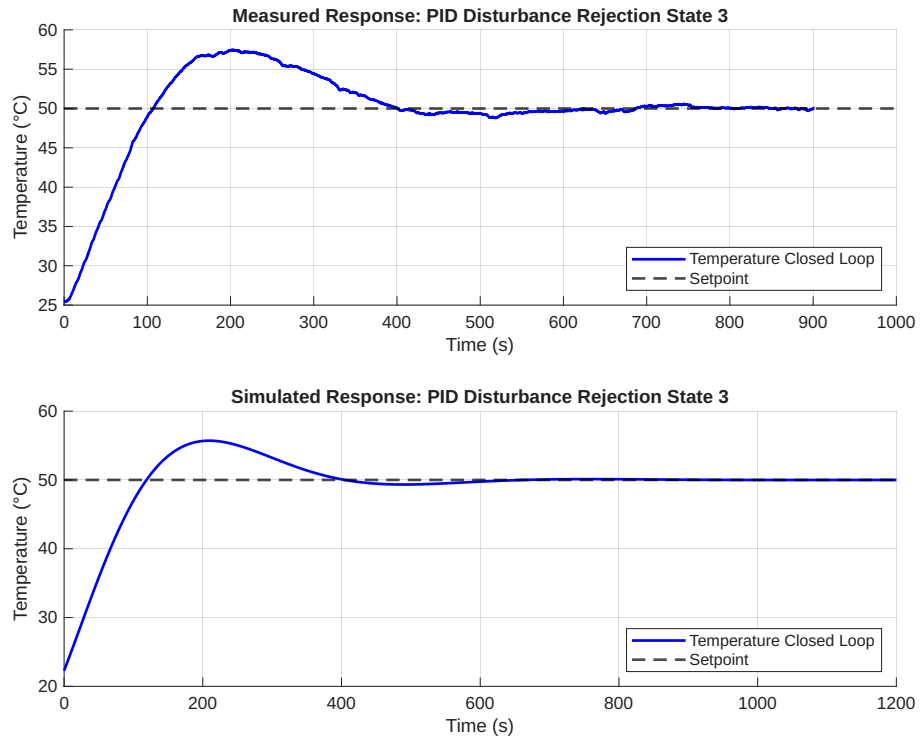


Figure 21: Measured Results for PID disturbance rejection tracking for state 3

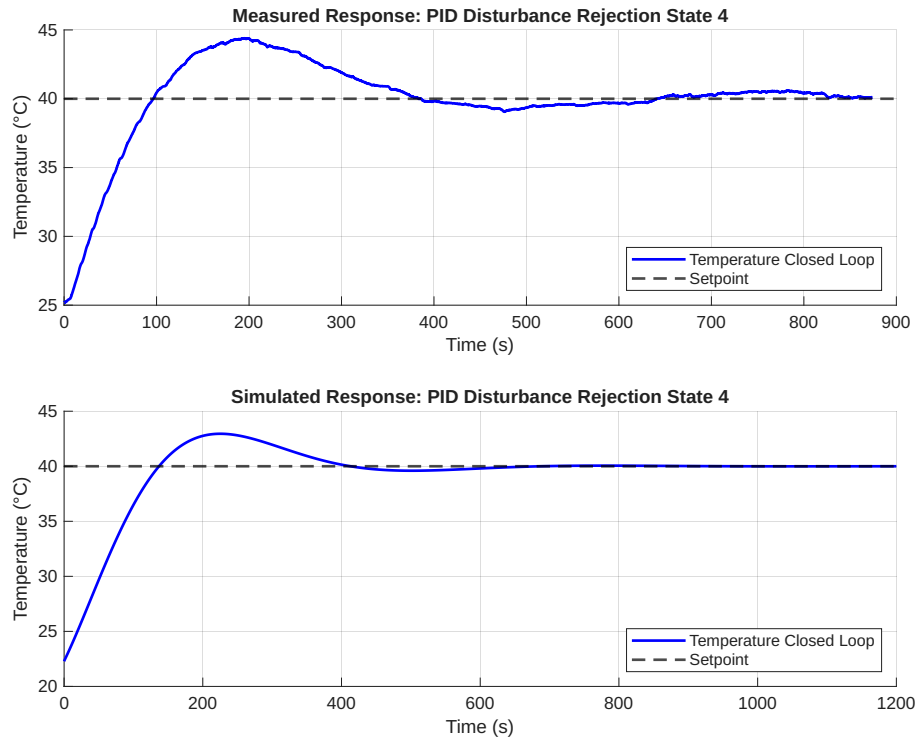


Figure 22: Measured Results for PID disturbance rejection tracking for state 4

5 Discussion

6 Code

6.1 Matlab Code

Listing 3: App Code For Sampling

```
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%% Auto Generated Code Start
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

classdef EBBPrac < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        UIFigure                matlab.ui.Figure
        GridLayout               matlab.ui.container.
            GridLayout
        LeftPanel                matlab.ui.container.
            Panel
        TabGroup                 matlab.ui.container.
            TabGroup
        Tab                      matlab.ui.container.
            Tab
        DataSaveEditField        matlab.ui.control.
            EditField
        DataSaveEditFieldLabel    matlab.ui.control.
            Label
        SamplingtoofastLabel      matlab.ui.control.
            Label
        UpdateChannelsButton      matlab.ui.control.
            Button
        StartButton               matlab.ui.control.
            StateButton
        StopButton                matlab.ui.control.
            StateButton
        OutputDropDown            matlab.ui.control.
            DropDown
        OutputDropDownLabel       matlab.ui.control.
            Label
        DutyCycleEditField        matlab.ui.control.
            NumericEditField
        DutyCycleEditFieldLabel    matlab.ui.control.
            Label
        DisturbanceEditField       matlab.ui.control.
```

```

        NumericEditField
DisturbanceEditFieldLabel    matlab.ui.control.
        Label
SetpointEditField            matlab.ui.control.
        NumericEditField
SetpointEditFieldLabel      matlab.ui.control.
        Label
DEditField                   matlab.ui.control.
        NumericEditField
DEditFieldLabel              matlab.ui.control.
        Label
IEditField                   matlab.ui.control.
        NumericEditField
IEditFieldLabel              matlab.ui.control.
        Label
PEditField                   matlab.ui.control.
        NumericEditField
PEditFieldLabel              matlab.ui.control.
        Label
Tab2                          matlab.ui.container.
        Tab
BaudRateEditField            matlab.ui.control.
        NumericEditField
BaudRateEditFieldLabel      matlab.ui.control.
        Label
COMPortEditField             matlab.ui.control.
        EditField
COMPortEditFieldLabel        matlab.ui.control.
        Label
RightPanel                   matlab.ui.container.
        Panel
UIAxes_2                     matlab.ui.control.
        UIAxes
UIAxes                        matlab.ui.control.
        UIAxes
end

% Properties that correspond to apps with auto-reflow
properties (Access = private)
    onePanelWidth = 576;
end
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Auto Generated Code End
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```

```

properties (Access = private)
    %Live lines 1 and 2 are for the temperature
    measurements.
    liveLine1
    liveLine2
    %Live lines 3 and 4 are for the PWM signals.
    liveLine3
    liveLine4
    P
    I
    D
    setpoint
    disturbance
    duty_cycle
    output
    capturing
    COM_port
    baud_rate
    window_size
    esp32
    data_save_name % Description
end

% Callbacks that handle component events
methods (Access = private)

    % Value changed function: PEditField
    function PEditFieldValueChanged(app, event)
        app.P = app.PEditField.Value;
    end

    % Value changed function: IEditField
    function IEditFieldValueChanged(app, event)
        app.I = app.IEditField.Value;
    end

    % Value changed function: DEditField
    function DEditFieldValueChanged(app, event)
        app.D = app.EDitField.Value;
    end

    % Value changed function: SetpointEditField
    function SetpointEditFieldValueChanged(app, event
    )
        app.setpoint = app.SetpointEditField.Value;

```

```

end

% Value changed function: DisturbanceEditField
function DisturbanceEditFieldValueChanged(app,
    event)
    app.disturbance = app.DisturbanceEditField.
        Value;
end

% Value changed function: DutyCycleEditField
function DutyCycleEditFieldValueChanged(app,
    event)
    app.duty_cycle = app.DutyCycleEditField.Value
        ;
end

% Value changed function: OutputDropDown
function OutputDropDownValueChanged(app, event)
    app.output = str2double(app.OutputDropDown.
        Value);
end

% Value changed function: StartButton
function StartButtonValueChanged(app, event)
    app.StartButton.Visible = "off";
    app.StopButton.Visible = "on";
    app.UpdateChannelsButton.Visible = "on";
    app.capturing = true;

    %Call all of the other functions to make sure
        their values are
    %captured.
    app.PEditFieldValueChanged();
    app.IEditFieldValueChanged();
    app.DEditFieldValueChanged();
    app.SetpointEditFieldValueChanged();
    app.DisturbanceEditFieldValueChanged();
    app.DutyCycleEditFieldValueChanged();
    app.OutputDropDownValueChanged();
    app.BaudRateEditFieldValueChanged();
    app.COMPortEditFieldValueChanged();
    app.DataSaveEditFieldValueChanged();

    if isempty(app.esp32)
        app.esp32 = serialport(app.COM_port, app.
            baud_rate);
        configureTerminator(app.esp32, "LF");
    end
end

```

```

flush(app.esp32);

pause(1);

%Make sure all info is read into esp.
writeline(app.esp32, sprintf("%d, %.2f", 0,
    app.P)); %
writeline(app.esp32, sprintf("%d, %.2f", 1,
    app.I)); %
writeline(app.esp32, sprintf("%d, %.2f", 2,
    app.D)); %
writeline(app.esp32, sprintf("%d, %.2f", 3,
    app.setpoint)); %
writeline(app.esp32, sprintf("%d, %.2f", 4,
    app.disturbance)); %
writeline(app.esp32, sprintf("%d, %d", 5, app
    .duty_cycle)); %
writeline(app.esp32, sprintf("%d, %.2f", 6,
    app.output)); %
writeline(app.esp32, sprintf("%d, %.2f", 7,
    app.capturing)); %

%Clear axes.
cla(app.UIAxes);
cla(app.UIAxes_2);

%Plotting axes for the temperature
measurements.
app.liveLine1 = animatedline(app.UIAxes, '
    Color', 'r');
app.liveLine2 = animatedline(app.UIAxes, '
    Color', 'b');

%Plotting axes for the PWM signals.
app.liveLine3 = animatedline(app.UIAxes_2, '
    Color', 'r');
app.liveLine4 = animatedline(app.UIAxes_2, '
    Color', 'b');

app.window_size = 30; %seconds

%Define data csv that we will save to.
fid = fopen(app.data_save_name + ".csv", 'wt'
    );

while(app.capturing)
    if app.esp32.NumBytesAvailable > 0
        rawData = readline(app.esp32);

```

```

data = str2double(split(rawData, ','));
data(5) = data(5)/1000;

if ~isnan(data(1)) && (length(data)
== 6)
    %Plot temperature data.
    addpoints(app.liveLine1, data(5),
        data(1));
    addpoints(app.liveLine2, data(5),
        data(2));
    xlim(app.UIAxes, [data(5) - app.
        window_size, data(5)]);

    %Plot PWM signals.
    addpoints(app.liveLine3, data(5),
        data(3));
    addpoints(app.liveLine4, data(5),
        data(4));
    xlim(app.UIAxes_2, [data(5) - app
        .window_size, data(5)]);

    %Write data to file.
    fprintf(fid, rawData);

    if data(end) == 0
        app.SamplingtoofastLabel.
            Visible = "on";
    else
        app.SamplingtoofastLabel.
            Visible = "off";
    end
end

drawnow limitrate
end

%Close file we are saving to.
fclose(fid);

%Send stop signal.
writeline(app.esp32, sprintf("%d, %.2f", 7,
    app.capturing));

clear app.esp32;
end

```

```

% Value changed function: COMPortEditField
function COMPortEditFieldValueChanged(app, event)
    app.COM_port = app.COMPortEditField.Value;
end

% Value changed function: BaudRateEditField
function BaudRateEditFieldValueChanged(app, event
    )
    app.baud_rate = app.BaudRateEditField.Value;
end

% Value changed function: StopButton
function StopButtonValueChanged(app, event)
    app.capturing = false;
    app.StopButton.Visible = "off";
    app.UpdateChannelsButton.Visible = "off";
    app.StartButton.Visible = "on";

end

% Button pushed function: UpdateChannelsButton
function UpdateChannelsButtonPushed(app, event)
    app.DutyCycleEditFieldValueChanged();
    app.OutputDropDownValueChanged();
    writeline(app.esp32, sprintf("%d, %.2f", 5,
        app.duty_cycle));
    writeline(app.esp32, sprintf("%d, %.2f", 6,
        app.output));
end

% Value changed function: DataSaveEditField
function DataSaveEditFieldValueChanged(app, event
    )
    app.data_save_name = app.DataSaveEditField.
        Value;

end

% Changes arrangement of the app based on
    UIFigure width
function updateAppLayout(app, event)
    currentFigureWidth = app.UIFigure.Position(3)
        ;
    if(currentFigureWidth <= app.onePanelWidth)
        % Change to a 2x1 grid
        app.GridLayout.RowHeight = {449, 449};
        app.GridLayout.ColumnWidth = {'1x'};
        app.RightPanel.Layout.Row = 2;
    end
end

```

```

        app.RightPanel.Layout.Column = 1;
    else
        % Change to a 1x2 grid
        app.GridLayout.RowHeight = {'1x'};
        app.GridLayout.ColumnWidth = {204, '1x'};
        app.RightPanel.Layout.Row = 1;
        app.RightPanel.Layout.Column = 2;
    end
end
end

%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Auto Generated Code Start
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create UIFigure and hide until all
        % components are created
        app.UIFigure = uifigure('Visible', 'off');
        app.UIFigure.AutoResizeChildren = 'off';
        app.UIFigure.Position = [100 100 702 449];
        app.UIFigure.Name = 'MATLAB App';
        app.UIFigure.SizeChangedFcn =
            createCallbackFcn(app, @updateAppLayout,
                true);

        % Create GridLayout
        app.GridLayout = uigridlayout(app.UIFigure);
        app.GridLayout.ColumnWidth = {204, '1x'};
        app.GridLayout.RowHeight = {'1x'};
        app.GridLayout.ColumnSpacing = 0;
        app.GridLayout.RowSpacing = 0;
        app.GridLayout.Padding = [0 0 0 0];
        app.GridLayout.Scrollable = 'on';

        % Create LeftPanel
        app.LeftPanel = uipanel(app.GridLayout);
        app.LeftPanel.Layout.Row = 1;

```

```

app.LeftPanel.Layout.Column = 1;

% Create TabGroup
app.TabGroup = uitabgroup(app.LeftPanel);
app.TabGroup.Position = [3 3 197 442];

% Create Tab
app.Tab = uitab(app.TabGroup);
app.Tab.Title = 'Tab';

% Create PEditFieldLabel
app.PEditFieldLabel = uilabel(app.Tab);
app.PEditFieldLabel.HorizontalAlignment = '
    right';
app.PEditFieldLabel.Position = [46 385 25
    22];
app.PEditFieldLabel.Text = 'P';

% Create PEditField
app.PEditField = uieditfield(app.Tab, '
    numeric');
app.PEditField.ValueChangedFcn =
    createCallbackFcn(app, @
        PEditFieldValueChanged, true);
app.PEditField.Position = [85 385 100 22];

% Create IEditFieldLabel
app.IEditFieldLabel = uilabel(app.Tab);
app.IEditFieldLabel.HorizontalAlignment = '
    right';
app.IEditFieldLabel.Position = [46 355 25
    22];
app.IEditFieldLabel.Text = 'I';

% Create IEditField
app.IEditField = uieditfield(app.Tab, '
    numeric');
app.IEditField.ValueChangedFcn =
    createCallbackFcn(app, @
        IEditFieldValueChanged, true);
app.IEditField.Position = [85 355 100 22];

% Create DEditFieldLabel
app.DEditFieldLabel = uilabel(app.Tab);
app.DEditFieldLabel.HorizontalAlignment = '
    right';
app.DEditFieldLabel.Position = [46 326 25
    22];

```

```

app.DEditFieldLabel.Text = 'D';

% Create DEditField
app.DEditField = uieditfield(app.Tab, '
    numeric');
app.DEditField.ValueChangedFcn =
    createCallbackFcn(app, @
        DEditFieldValueChanged, true);
app.DEditField.Position = [85 326 100 22];

% Create SetpointEditFieldLabel
app.SetpointEditFieldLabel = uilabel(app.Tab)
    ;
app.SetpointEditFieldLabel.
    HorizontalAlignment = 'right';
app.SetpointEditFieldLabel.Position = [21 296
    49 22];
app.SetpointEditFieldLabel.Text = 'Setpoint';

% Create SetpointEditField
app.SetpointEditField = uieditfield(app.Tab,
    'numeric');
app.SetpointEditField.ValueChangedFcn =
    createCallbackFcn(app, @
        SetpointEditFieldValueChanged, true);
app.SetpointEditField.Position = [85 296 100
    22];

% Create DisturbanceEditFieldLabel
app.DisturbanceEditFieldLabel = uilabel(app.
    Tab);
app.DisturbanceEditFieldLabel.
    HorizontalAlignment = 'right';
app.DisturbanceEditFieldLabel.Position = [1
    265 69 22];
app.DisturbanceEditFieldLabel.Text = '
    Disturbance';

% Create DisturbanceEditField
app.DisturbanceEditField = uieditfield(app.
    Tab, 'numeric');
app.DisturbanceEditField.ValueChangedFcn =
    createCallbackFcn(app, @
        DisturbanceEditFieldValueChanged, true);
app.DisturbanceEditField.Position = [85 265
    100 22];

% Create DutyCycleEditFieldLabel

```

```

app.DutyCycleEditFieldLabel = uilabel(app.Tab
    );
app.DutyCycleEditFieldLabel.
    HorizontalAlignment = 'right';
app.DutyCycleEditFieldLabel.Position = [9 235
    63 22];
app.DutyCycleEditFieldLabel.Text = 'Duty
    Cycle';

% Create DutyCycleEditField
app.DutyCycleEditField = uieditfield(app.Tab,
    'numeric');
app.DutyCycleEditField.ValueChangedFcn =
    createCallbackFcn(app, @
        DutyCycleEditFieldValueChanged, true);
app.DutyCycleEditField.Position = [86 235 100
    22];
app.DutyCycleEditField.Value = 100;

% Create OutputDropDownLabel
app.OutputDropDownLabel = uilabel(app.Tab);
app.OutputDropDownLabel.HorizontalAlignment =
    'right';
app.OutputDropDownLabel.Position = [30 199 41
    22];
app.OutputDropDownLabel.Text = 'Output';

% Create OutputDropDown
app.OutputDropDown = uidropdown(app.Tab);
app.OutputDropDown.Items = {'None', 'Channel1
    ', 'Channel2', 'Both'};
app.OutputDropDown.ItemsData = {'3', '0', '1'
    , '2'};
app.OutputDropDown.ValueChangedFcn =
    createCallbackFcn(app, @
        OutputDropDownValueChanged, true);
app.OutputDropDown.Position = [85 199 100
    22];
app.OutputDropDown.Value = '3';

% Create StopButton
app.StopButton = uibutton(app.Tab, 'state');
app.StopButton.ValueChangedFcn =
    createCallbackFcn(app, @
        StopButtonValueChanged, true);
app.StopButton.Visible = 'off';
app.StopButton.Text = 'Stop';
app.StopButton.Position = [85 168 100 23];

```

```

% Create StartButton
app.StartButton = uibutton(app.Tab, 'state');
app.StartButton.ValueChangedFcn =
    createCallbackFcn(app, @
        StartButtonValueChanged, true);
app.StartButton.Text = 'Start';
app.StartButton.Position = [86 168 100 23];

% Create UpdateChannelsButton
app.UpdateChannelsButton = uibutton(app.Tab,
    'push');
app.UpdateChannelsButton.ButtonPushedFcn =
    createCallbackFcn(app, @
        UpdateChannelsButtonPushed, true);
app.UpdateChannelsButton.Visible = 'off';
app.UpdateChannelsButton.Position = [81 136
    108 23];
app.UpdateChannelsButton.Text = 'Update
    Channels';

% Create SamplingtoofastLabel
app.SamplingtoofastLabel = uilabel(app.Tab);
app.SamplingtoofastLabel.Visible = 'off';
app.SamplingtoofastLabel.Position = [90 12
    101 22];
app.SamplingtoofastLabel.Text = 'Sampling too
    fast!';

% Create DataSaveEditFieldLabel
app.DataSaveEditFieldLabel = uilabel(app.Tab)
;
app.DataSaveEditFieldLabel.
    HorizontalAlignment = 'right';
app.DataSaveEditFieldLabel.Position = [10 96
    61 22];
app.DataSaveEditFieldLabel.Text = 'Data Save'
;

% Create DataSaveEditField
app.DataSaveEditField = uieditfield(app.Tab,
    'text');
app.DataSaveEditField.ValueChangedFcn =
    createCallbackFcn(app, @
        DataSaveEditFieldValueChanged, true);
app.DataSaveEditField.Position = [86 96 100
    22];
app.DataSaveEditField.Value = 'PWM1StepV0';

```

```

% Create Tab2
app.Tab2 = uitab(app.TabGroup);
app.Tab2.Title = 'Tab2';

% Create COMPortEditFieldLabel
app.COMPortEditFieldLabel = uilabel(app.Tab2);
;
app.COMPortEditFieldLabel.HorizontalAlignment = 'right';
app.COMPortEditFieldLabel.Position = [13 385 58 22];
app.COMPortEditFieldLabel.Text = 'COM Port';

% Create COMPortEditField
app.COMPortEditField = uieditfield(app.Tab2, 'text');
app.COMPortEditField.ValueChangedFcn = createCallbackFcn(app, @COMPortEditFieldValueChanged, true);
app.COMPortEditField.Position = [85 385 100 22];
app.COMPortEditField.Value = '/dev/ttyUSB0';

% Create BaudRateEditFieldLabel
app.BaudRateEditFieldLabel = uilabel(app.Tab2);
;
app.BaudRateEditFieldLabel.HorizontalAlignment = 'right';
app.BaudRateEditFieldLabel.Position = [9 355 62 22];
app.BaudRateEditFieldLabel.Text = 'Baud Rate';
;

% Create BaudRateEditField
app.BaudRateEditField = uieditfield(app.Tab2, 'numeric');
app.BaudRateEditField.ValueChangedFcn = createCallbackFcn(app, @BaudRateEditFieldValueChanged, true);
app.BaudRateEditField.Position = [85 355 100 22];
app.BaudRateEditField.Value = 115200;

% Create RightPanel
app.RightPanel = uipanel(app.GridLayout);
app.RightPanel.Layout.Row = 1;
app.RightPanel.Layout.Column = 2;

```

```

        % Create UIAxes
        app.UIAxes = uiaxes(app.RightPanel);
        title(app.UIAxes, 'Title')
        xlabel(app.UIAxes, 'X')
        ylabel(app.UIAxes, 'Y')
        zlabel(app.UIAxes, 'Z')
        app.UIAxes.Position = [7 213 489 235];

        % Create UIAxes_2
        app.UIAxes_2 = uiaxes(app.RightPanel);
        title(app.UIAxes_2, 'Title')
        xlabel(app.UIAxes_2, 'X')
        ylabel(app.UIAxes_2, 'Y')
        zlabel(app.UIAxes_2, 'Z')
        app.UIAxes_2.Position = [8 7 488 207];

        % Show the figure after all components are
        % created
        app.UIFigure.Visible = 'on';
    end
end

% App creation and deletion
methods (Access = public)

    % Construct app
    function app = EBBPrac

        % Create UIFigure and components
        createComponents(app)

        % Register the app with App Designer
        registerApp(app, app.UIFigure)

        if nargin == 0
            clear app
        end
    end

    % Code that executes before app deletion
    function delete(app)

        % Delete UIFigure when app is deleted
        delete(app.UIFigure)
    end
end
end
end

```

```
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Auto Generated Code end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
```

6.2 Model Fit Code

Listing 4: Model fitting and plotting code

```
% Obtaining the transfer function
% Package the input arrays of both transistors into one
  matrix (Needs to be columns)
PWM1StepV1Data = readmatrix('..\Data1\Step_PWM1_V0.csv');
PWM1StepV1 = PWM1StepV1Data(7920:end,1:5);
PWM2StepV0Data = readmatrix('..\Data1\Step_PWM2_V0.csv');
PWM2StepV0 = PWM2StepV0Data(6400:end,1:5);

PWM1TestDataFile = readmatrix('..\Data1\
  Step_PWM1_VtestData50.csv');
PWM1TestData = PWM1TestDataFile(6230:end,1:5);
PWM2TestDataFile = readmatrix('..\Data1\
  Step_PWM2_VtestData50.csv');
PWM2TestData = PWM2TestDataFile(6160:end,1:5);

% pwm1 data prep
V_PWM1 = [PWM1StepV1(:,3)];
Temp_PWM1 = [PWM1StepV1(:,1)-22.28, PWM1StepV1(:,2)
  -21.24];
time_step = 0.005;

% pwm2 data prep
V_PWM2 = [PWM2StepV0(:,4)];
Temp_PWM2 = [PWM2StepV0(:,1)-20.26, PWM2StepV0(:,2)
  -19.05];

% pwm1 test data prep
V_PWM1_Test = [PWM1TestData(:,3)];
Temp_PWM1_test = [PWM1TestData(:,1)-21.2, PWM1TestData
  (:,2)-20.2];

% pwm2 test data prep
V_PWM2_Test = [PWM2TestData(:,4)];
Temp_PWM2_test = [PWM2TestData(:,1)-21.1, PWM2TestData
  (:,2)-20.2] ;
```

```

% use iddata to set up the data for tfest
tfest_data1 = iddata(Temp_PWM1, V_PWM1, time_step);
tfest_data2 = iddata(Temp_PWM2, V_PWM2, time_step);
testData1 = iddata(Temp_PWM1_test, V_PWM1_Test, time_step
);
testData2 = iddata(Temp_PWM2_test, V_PWM2_Test, time_step
);

% estimate nr of poles and zeros
n_poles = 1;
n_zeros = 0;

% Estimate the transfer functions independently (SISO) to
    avoid idss error
% tfest_data(samples, output, input)
sys11 = tfest(tfest_data1(:, 1, 1), n_poles, n_zeros, NaN
); % PWM1 to T1
sys21 = tfest(tfest_data1(:, 2, 1), n_poles, n_zeros, NaN
); % PWM1 to T2

sys12 = tfest(tfest_data2(:, 1, 1), n_poles, n_zeros, NaN
); % PWM2 to T1
sys22 = tfest(tfest_data2(:, 2, 1), n_poles, n_zeros, NaN
); % PWM2 to T2

% Combine the individual SISO models back into SIMO
    systems
simo_transfer1 = [sys11; sys21];
simo_transfer2 = [sys12; sys22];

% set up graph details
tfest_data1.Name = 'Measured Data';
tfest_data2.Name = 'Measured Data';
testData1.Name = 'Measured Data';
testData2.Name = 'Measured Data';
simo_transfer2.Name = 'Simulated Model';
simo_transfer1.Name = 'Simulated Model';

figure(1)
compare(tfest_data1, simo_transfer1);
legend('Location', 'best');
ylabel('Change in Temperature (C)')

figure(2)
compare(tfest_data2, simo_transfer2);
legend('Location', 'best');
ylabel('Change in Temperature (C)')

```

```

figure(3)
compare(testData1, simo_transfer1);
legend('Location', 'best');
ylabel('Change in Temperature (C)')

figure(4)
compare(testData2, simo_transfer2);
legend('Location', 'best');
ylabel('Change in Temperature (C)')

lines = findobj(gcf, 'Type', 'Line');
set(lines, 'LineWidth', 1.5);

% display the transfer functions of PWM1
[num1, den1] = tfdata(simo_transfer1(1, 1), 'v');
delay1 = simo_transfer1(1, 1).IODelay; % Extract dead
    time
[num2, den2] = tfdata(simo_transfer1(2, 1), 'v');
delay2 = simo_transfer1(2, 1).IODelay;

disp('--- Transfer Function 1 (PWM1 to T1) ---');
disp('Numerator:'); disp(num1);
disp('Denominator:'); disp(den1);
disp(['Estimated Dead Time (s): ', num2str(delay1)]);

disp('--- Transfer Function 2 (PWM1 to T2) ---');
disp('Numerator:'); disp(num2);
disp('Denominator:'); disp(den2);
disp(['Estimated Dead Time (s): ', num2str(delay2)]);

% display transfer functions of PWM2
[num12, den12] = tfdata(simo_transfer2(1, 1), 'v');
delay12 = simo_transfer2(1, 1).IODelay;
[num22, den22] = tfdata(simo_transfer2(2, 1), 'v');
delay22 = simo_transfer2(2, 1).IODelay;

disp('--- Transfer Function 1 (PWM2 to T1) ---');
disp('Numerator:'); disp(num12);
disp('Denominator:'); disp(den12);
disp(['Estimated Dead Time (s): ', num2str(delay12)]);

disp('--- Transfer Function 2 (PWM2 to T2) ---');
disp('Numerator:'); disp(num22);
disp('Denominator:'); disp(den22);
disp(['Estimated Dead Time (s): ', num2str(delay22)]);

```

6.3 Prac 2 — Controller Design (P and PI)

6.4 Prac 2 — Report Figure Generation (P and PI)

6.5 MCU Code

```
#include <fmt/core.h>
#include <Arduino.h>
#include <iostream>
#include <sstream>
#include <vector>
#include <string>
#include <queue>

double P = 0;
double I = 0;
double D = 0;
double setpoint = 0;
double disturbance = 0;
double duty = 128; //50% duty cycle by default.
int control_outputs = 0; //By default only pwm1 is on.
bool start = 0; //Start off.
int message = 100; //Flags for certain messages.
int start_time = 0;
bool startControlling = 0; //See if we should start
    controlling, or continue just measuring.

//PID variables
float error = 0;
float integral = 0;
float previous_error = 0;
float derivative = 0;
bool do_integration = true;

double temp1 = 0;
double temp2 = 0;
int N_samples = 200;
std::queue<double> samples1;
std::queue<double> samples2;

hw_timer_t *timer = NULL;
const int PWM_Out_1 = GPIO_NUM_32;
const int PWM_Out_2 = GPIO_NUM_25;
const int PWM_Meas_1 = GPIO_NUM_33;
const int PWM_Meas_2 = GPIO_NUM_26;

const int Temp_Meas1 = GPIO_NUM_27;
```

```

const int Temp_Meas2 = GPIO_NUM_35;

const double A = (3.3/4095.0);
const int PWM_freq = 10;
const int PWM_res = 8;

const int BaudRate = 1152000;
const int sampling_period = 5000; //In microseconds.
const float dt = sampling_period/1000000.0;
bool first_run = true;

void setup() {
    //Start the USB-C serial/UART port.
    Serial.begin(BaudRate);

    //Setup I/O pins.
    pinMode(Temp_Meas1, INPUT);
    pinMode(Temp_Meas2, INPUT);
    pinMode(PWM_Meas_1, INPUT);
    pinMode(PWM_Meas_2, INPUT);

    ledcChangeFrequency(0, PWM_freq, PWM_res);
    ledcChangeFrequency(1, PWM_freq, PWM_res);
    ledcAttachPin (PWM_Out_1, 0);
    ledcAttachPin(PWM_Out_2, 1);

    timer = timerBegin(0, 80, true);

    samples1.push(0);
    samples2.push(0);
}

void loop() {

    //If receiving communication from computer, update values.
    if(Serial.available() > 0)
    {
        String received = Serial.readStringUntil('\n');
        received.trim();

        std::istringstream stream(received.c_str());
        std::string command;
        std::string value_str;
        double value;

        std::getline(stream, command, ',');
        std::getline(stream, value_str, ',');
        value = std::stod(value_str);
    }
}

```

```

    if(command == "0")
    {
        P = value;
    }
    else if(command == "1")
    {
        I = value;
    }
    else if(command == "2")
    {
        D = value;
    }
    else if(command == "3")
    {
        setpoint = value;
    }
    else if(command == "4")
    {
        disturbance = value;
        disturbance = (pow(2, PWM_res) - 1)*(disturbance
            /100.0);
    }
    else if(command == "5") //Change from duty cycle set to
        the startControlling flag.
    {
        startControlling = value;
    }
    else if(command == "6")
    {
        control_outputs = value;
        //Serial.print("Igot");
        //Serial.println(control_outputs);
    }
    else if(command == "7")
    {
        start = value;

        //Check if start is true, then start timer.
        if(start)
        {
            start_time = millis();
        }
    }
}

//If start flag is true, start measuring inputs/start
controlling aswell.

```

```

if(start)
{
    //Check when we started in order to see how long we were
    busy.
    timerRestart(timer);

    if(startControlling)
    {
        //Turn on the appropriate PWM signals.
        ledcWrite(0, duty);
        ledcWrite(1, disturbance);
    }

    //Interpret sensor data.
    double temp1_raw = (analogReadMilliVolts(Temp_Meas1)
        /1000.0 - 0.5)/0.01;
    double temp2_raw = (analogReadMilliVolts(Temp_Meas2)
        /1000.0 - 0.5)/0.01;

    if(samples1.size() > N_samples)
    {
        temp1 -= samples1.front();
        temp2 -= samples2.front();

        samples1.pop();
        samples2.pop();
    }

    temp1 += temp1_raw/N_samples;
    temp2 += temp2_raw/N_samples;

    samples1.push(temp1_raw/N_samples);
    samples2.push(temp2_raw/N_samples);

    //Send sensor input data as a comma seperated list.
    std::string sending = fmt::format("{0},{1},{2},{3},{4},{5}",
        temp1, temp2, analogRead(PWM_Meas_1)*A, analogRead(
            PWM_Meas_2)*A, millis() - start_time, message);

    Serial.println(String(sending.c_str()));
    Serial.flush();

    if (startControlling)
    {
        //PID controller start
        previous_error = error;
    }
}

```

```

error = setpoint - templ;

// Prevent derivative kick on first run

if (first_run) {
    previous_error = error;
    first_run = false;
}

float next_integral = integral + error * dt;

derivative = (error - previous_error)/dt;

float pid_out = P * error + I*next_integral + D*
    derivative;
//anti-windup logic
if (pid_out >= 3.3 )
{
    duty = 1;
    if ( error < 0){
        integral = next_integral;
    }
}

else if (pid_out <= 0.0 ) {
    duty = 0.0; // You cannot have a negative duty cycle
                for a simple heater/cooler
    if ( error > 0){
        integral = next_integral;
    }
}

else{
    duty = pid_out/3.3;
    integral = next_integral;
}
duty = (pow(2, PWM_res) - 1)*(duty);

// std::string sending = fmt::format("{0}, {1}, {2},
// {3}, {4}", P, I, D, error, duty);
// Serial.println(String(sending.c_str()));
// Serial.flush();
}
//Otherwise turn off control signals.
else
{

```

```

    ledcWrite(0, 0);
    ledcWrite(1, 0);
}

//Check how long everything took in microseconds.
int elapsed = timerReadMicros(timer);

if(elapsed <= sampling_period)
{
    delayMicroseconds(sampling_period - elapsed);
    message = 100; //Reset message flag to show nothing.
}
else
{
    delayMicroseconds(sampling_period);
    message = 0; //Set message flag for sampling too fast.
}

//Otherwise deactivate PWM outputs.
}
else
{
    ledcWrite(0, 0);
    ledcWrite(1, 0);
}
}

```

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